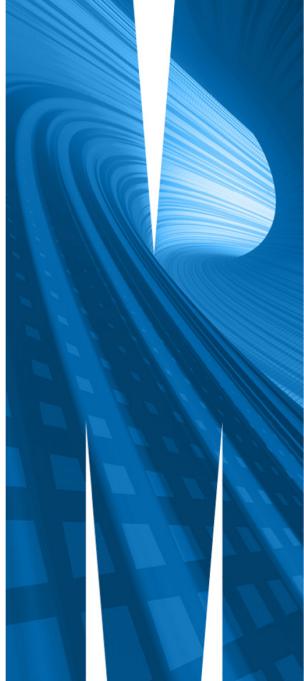


ETX1100/ETX5900

Business Statistics

Week 6: Multiple regression and model
evaluation

12 January 2027



Outline

- 1 From one predictor to several
- 2 Building the model in Excel
- 3 Inference and model evaluation
- 4 Categorical predictors
- 5 Refinement and diagnostics

Today's journey

- 1 Extend regression to several predictors.
- 2 Interpret partial slopes while holding other variables constant.
- 3 Test coefficients and evaluate overall model fit.
- 4 Represent categorical predictors with dummy variables.
- 5 Refine, diagnose, predict, and communicate a model.

Outline

- 1 From one predictor to several
- 2 Building the model in Excel
- 3 Inference and model evaluation
- 4 Categorical predictors
- 5 Refinement and diagnostics

Retrieval check

- What does a simple-regression slope mean?
- What does R^2 measure?
- Why inspect residuals?
- Why can an omitted variable distort a simple-regression relationship?

Multiple linear regression

Population model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki} + \varepsilon_i.$$

Fitted model:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \dots + \hat{\beta}_k X_{ki}.$$

Multiple regression estimates each predictor's association with Y conditional on the others in the model.

Partial slopes

$\hat{\beta}_j$ is the predicted mean change in Y for a one-unit increase in X_j , **holding all other included predictors constant.**

This is not automatically a causal effect. “Holding constant” is a model comparison, not necessarily an intervention.

Watch me: simple versus multiple slopes

In real-estate data, the simple slope for area mixes area with correlated features such as bedrooms and bathrooms.

Add those predictors and compare:

- the area coefficient;
- its standard error; and
- the meaning of the conditional comparison.

A changed coefficient is evidence that model context matters.

Now you try: interpret partial slopes

Time allowed: 6 minutes

For

$$\widehat{Price} = 86.8 + 47.4Beds + 216.5Baths + 14.5Cars + 0.831Area,$$

interpret the coefficients of Baths and Area, including units and the variables held constant. Explain why neither wording should use “causes”.

Outline

- 1 From one predictor to several
- 2 Building the model in Excel
- 3 Inference and model evaluation
- 4 Categorical predictors
- 5 Refinement and diagnostics

Prepare the data

- One row per observation.
- One column per variable.
- Response in one contiguous column.
- Predictors in adjacent columns for Excel's Regression tool.
- Clear headers and consistent units.
- Missing values resolved explicitly—not silently converted to zero.

Watch me: fit a real-estate model

Open week6 - realestate - data.xlsx.

- 1 Choose Price as the Input Y Range.
- 2 Select a contiguous block of numerical predictors for Input X Range.
- 3 Tick **Labels**, residuals, and residual plots.
- 4 Place output on a new worksheet.
- 5 Map every output column back to the model equation.

Reading Excel regression output

Three regions answer different questions:

- **Regression Statistics:** fit and typical error.
- **ANOVA:** whether the predictor set has overall linear explanatory value.
- **Coefficients:** estimated partial slopes, uncertainty, tests, and intervals.

Never interpret a p-value before confirming which row and hypothesis it belongs to.

Now you try: reconstruct the model

Time allowed: 7 minutes

From your Excel output:

- 1 write the fitted equation;
- 2 interpret two slopes;
- 3 identify any meaningless intercept interpretation;
- 4 predict price for one observed property; and
- 5 calculate its residual.

Outline

- 1 From one predictor to several
- 2 Building the model in Excel
- 3 Inference and model evaluation**
- 4 Categorical predictors
- 5 Refinement and diagnostics

Test an individual coefficient

For predictor X_j :

$$H_0 : \beta_j = 0, \quad H_1 : \beta_j \neq 0.$$

Excel's coefficient p-value is two-sided. A small p-value indicates evidence of a linear association with Y after accounting for other included predictors.

Watch me: coefficient evidence

For each coefficient:

- 1 identify estimate and standard error;
- 2 verify $t = \hat{\beta}_j / SE(\hat{\beta}_j)$;
- 3 compare p-value with α ;
- 4 inspect the confidence interval; and
- 5 state the conditional business conclusion.

Now you try: decide what contributes

Time allowed: 7 minutes

At $\alpha = 0.05$:

- 1 identify predictors with evidence of contribution;
- 2 identify candidates for removal;
- 3 explain why p-value alone is not enough to delete a variable; and
- 4 name one variable that business logic might require even if its p-value is large.

R^2 and adjusted R^2

- R^2 cannot decrease when predictors are added.
- Adjusted R^2 penalises additional predictors that add little explanatory value.
- Compare models fitted to the same response and observations.
- A high value does not establish correct specification, causality, or good out-of-sample prediction.

Now you try: compare two models

Time allowed: 6 minutes

Fit a full and a reduced real-estate model.

Compare:

- 1 adjusted R^2 ;
- 2 standard error of regression;
- 3 coefficient stability;
- 4 residual behaviour; and
- 5 interpretability.

Recommend one model and justify the trade-off.

Overall F test

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$$

H_1 : at least one slope is non-zero.

Excel reports the p-value as **Significance F**. This evaluates the predictor set collectively; it does not say every predictor is significant.

Watch me: separate model and coefficient claims

A small Significance F supports an overall linear relationship between Y and the predictor set.

Then inspect individual coefficient tests to learn which predictors contribute conditionally. A significant model can contain non-significant individual predictors.

Now you try: interpret the ANOVA block

Time allowed: 5 minutes

Using your model output:

- 1 state the F -test hypotheses;
- 2 report Significance F ;
- 3 make the decision at 5%;
- 4 write the contextual conclusion; and
- 5 explain why it does not prove every slope is non-zero.

Outline

- 1 From one predictor to several
- 2 Building the model in Excel
- 3 Inference and model evaluation
- 4 Categorical predictors**
- 5 Refinement and diagnostics

Dummy variables

For a categorical predictor with j categories, use $j - 1$ dummy variables.

- Each dummy takes 0 or 1.
- The omitted category is the **base category**.
- Dummy coefficients compare their category with the base, holding other predictors constant.
- Including all j dummies with an intercept creates perfect redundancy.

Watch me: create training-method dummies

Open `week6-underwriting-data.xlsx`.

With `courseware app` as the base category, create:

- `DClassroom = IF(Method="Classroom",1,0);`
- `DOnline = IF(Method="Online",1,0).`

Check that `courseware` rows have zero for both dummies.

Now you try: verify the coding

Time allowed: 5 minutes

- 1 Complete both dummy columns.
- 2 Check at least two observations from each training method.
- 3 Explain why no DCourseware column enters the model.
- 4 Predict the dummy values for a new online trainee.

Interpret dummy coefficients

For

$$\widehat{Exam} = -63.981 + 1.126Proficiency - 22.289DClassroom + 8.088DOnline$$

- classroom is predicted 22.289 points below courseware at equal proficiency;
- online is predicted 8.088 points above courseware at equal proficiency.

The base category is essential to the interpretation.

Now you try: underwriting model

Time allowed: 10 minutes

Fit the underwriting model and:

- 1 interpret the proficiency slope;
- 2 interpret both method coefficients;
- 3 predict the score for proficiency 100 under each method;
- 4 test each coefficient at 5%; and
- 5 interpret adjusted R^2 and Significance F.

Outline

- 1 From one predictor to several
- 2 Building the model in Excel
- 3 Inference and model evaluation
- 4 Categorical predictors
- 5 Refinement and diagnostics

Model reduction is an argument

A defensible reduction process considers:

- the business question;
- coefficient evidence and intervals;
- adjusted R^2 and predictive error;
- coefficient stability after refitting;
- hierarchy and important controls; and
- residual diagnostics.

Do not repeatedly delete the largest p-value without explanation.

Residual diagnostics

Look for:

- curvature: missing nonlinear structure;
- funnel shape: non-constant variance;
- clusters: missing groups;
- extreme residuals: poor fit for particular observations;
and
- influential points: observations that may strongly affect coefficients.

Watch me: refit and compare

Remove one defensible candidate from the full real-estate model, refit, and compare:

- remaining coefficient estimates;
- adjusted R^2 ;
- standard error;
- residual pattern; and
- clarity of interpretation.

Record the reason for the change before viewing whether fit “improves”.

Now you try: model recommendation

Time allowed: 10 minutes

Recommend either the full or reduced model.

Your brief must include:

- 1 fitted equation;
- 2 two coefficient interpretations;
- 3 overall and individual evidence;
- 4 adjusted R^2 and residual assessment;
- 5 one prediction; and
- 6 two limitations.

Common traps

- Interpreting a partial slope as a simple bivariate association.
- Forgetting “holding other included predictors constant”.
- Treating the base category as missing data.
- Interpreting Significance F as proof every predictor matters.
- Comparing R^2 across different responses or samples.
- Deleting variables mechanically by p-value.
- Claiming causal effects from an observational regression.

Closing check

Can you:

- build a multiple regression in Excel?
- interpret numerical and dummy-variable slopes conditionally?
- test individual coefficients and the overall model?
- compare models using adjusted R^2 , error, and diagnostics?
- make predictions with correct dummy coding?
- recommend a model with evidence and limitations?